

Biology - Expt 4

Estimation of Glucose Concentration using GOD-POD Assay

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Abstract

The GOD-POD (Glucose Oxidase-Peroxidase) assay is a widely used enzymatic method for the quantitative determination of glucose concentration in biological samples. This experiment aims to estimate the glucose concentration in a given sample by utilizing the GOD-POD assay, which involves the enzymatic oxidation of glucose to produce hydrogen peroxide, followed by a colorimetric reaction that can be measured spectrophotometrically. By comparing the absorbance of the sample to a standard curve generated from known glucose concentrations, we can accurately determine the glucose content in the sample.

Aim

To estimate the glucose concentration in a given sample using the GOD-POD assay.

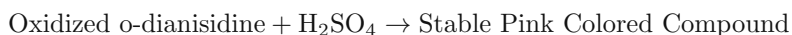
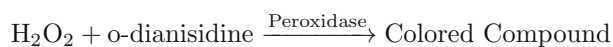
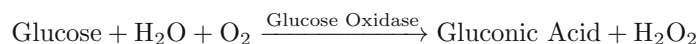
Apparatus

- GOD-POD Reagent Kit
- Standard Glucose Solution
- Test Tubes
- Micropipettes and Tips
- Spectrophotometer
- Cuvettes
- Distilled Water

Working Principle

Glucose oxidase catalyzes the oxidation of glucose to gluconic acid and hydrogen peroxide. The hydrogen peroxide produced then reacts with a chromogenic substrate (o-dianisidine) in the presence of peroxidase to form a brown colored compound. This further reacts with H_2SO_4 to produce a stable pink coloured compound. The intensity of the color measured at 540 nm is directly proportional to the glucose concentration in the sample. By measuring the absorbance of the colored compound at a specific wavelength, we can determine the glucose concentration by comparing it to a standard curve generated from known glucose concentrations.

The Glucose oxidase enzyme is an oxido reductase that specifically catalyzes the oxidation of glucose to D-glucono-1,5-lactone and hydrogen peroxide. The overall reaction can be summarized as follows:



Observations and Results

Concentration of Glucose solution given = 200 μ g/mL to prepare standards.

The absorbance values for different standard glucose concentrations and the sample are as follows:

Sample Number	Required conc for preparation of standard (μ g/mL)	Required Volume of Glucose (μ L)
1A, 1B, 1C	5	12.5
2A, 2B, 2C	10	25
3A, 3B, 3C	15	37.5
4A, 4B, 4C	20	50
5A, 5B, 5C	25	62.5

Figure 1: Required volume of glucose solution for preparation of standards

Sample	Absorbance at 540 nm
1A	0.130
1B	0.136
1C	0.136
2A	0.229
2B	0.239
2C	0.251
3A	0.384
3B	0.390
3C	0.372
4A	0.494
4B	0.485
4C	0.490
5A	0.627
5B	0.602
5C	0.619
Unknown (10)	0.243
Unknown (10)	0.234
Unknown (10)	0.215

Figure 2: Absorbance values for standards and sample

We plot a standard curve of absorbance vs glucose concentration and use it to determine the glucose concentration in the sample based on its absorbance value.

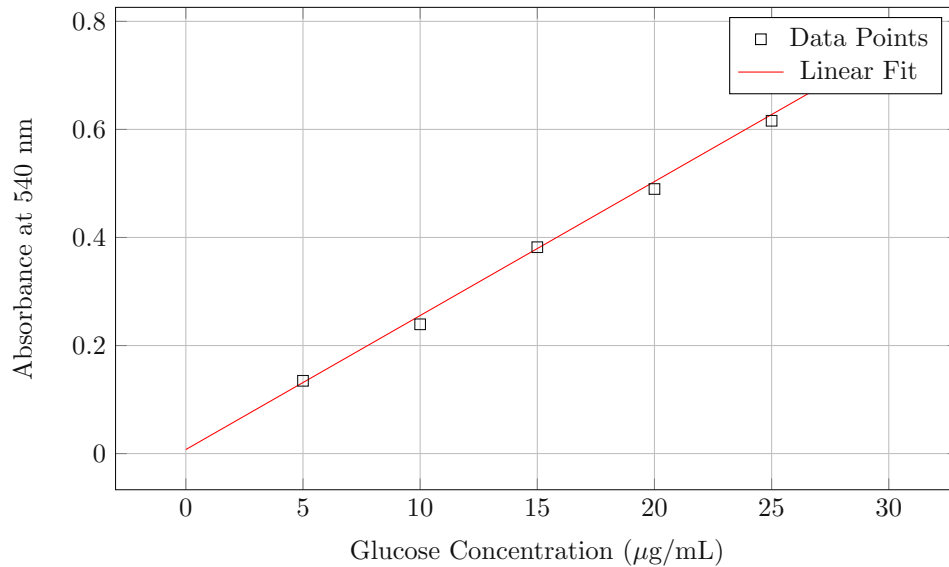


Figure 3: Curve of Absorbance vs Glucose Concentration with Linear Fit ($m = 0.0248$, $c = 0.0075$)

Calculating Glucose Concentration in Unknown Sample

From the linear fit equation of the standard curve, we can calculate the glucose concentration in the unknown sample using its absorbance value.

For Unknown Sample with average absorbance = $\frac{0.243+0.234+0.215}{3} = 0.2307$

Using the linear fit equation:

$$\text{Absorbance} = m \times \text{Concentration} + c$$

$$0.2307 = 0.0248 \times \text{Concentration} + 0.0075$$

$$\text{Concentration} = \frac{0.2307 - 0.0075}{0.0248} \approx 9.05 \mu\text{g/mL}$$

Conclusion

The glucose concentration in the unknown sample was estimated using the GOD-POD assay and found to be approximately $9.05 \mu\text{g/mL}$ based on the absorbance measurement and the standard curve generated from known glucose concentrations.